CS 03500 - Foundations of Cybersecurity

Project Part 1:

Data Classification Standards and Risk Assessment Methodology

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Data Classification Standards

Data classification involves sorting data by type, sensitivity, and importance to facilitate efficient management and compliance with legal and regulatory standards. This practice is crucial for organizations looking to strengthen data security, improve operations, and enhance data accessibility.

Data classification standards applicable to a company like Fullsoft should include data labeling levels such as:

- Restricted data: Restricted data typically pertains to information that can only be accessed by authorized personnel. This classification ensures that sensitive information remains secure and minimizes the risk of unauthorized disclosures.
- Confidential data: Confidential data refers to information or data owned by the organization, such as intellectual property, that requires clearance to be granted for access.
- Internal data: Internal data refers to information or data shared within an organization, and these communications are not meant to be disclosed outside the organization.
- Public data: This data is available to the public, both locally and online. It can be freely accessed by anyone without restrictions or special permissions.

Risk assessment methodologies:

NIST SP 800-30, Guide for Conducting Risk Assessments

NIST SP 800-30 offers a systematic approach to performing information security risk assessments, covering the processes of identifying, analyzing, and evaluating risks. During the assessment, it focuses on identifying assets, threats, vulnerabilities, controls, and the likelihood of risks materializing.

Operationally Critical Threat, Asset, and Vulnerability Evaluation (OCTAVE), Allegro version

OCTAVE Allegro is an efficient and streamlined risk assessment methodology that allows organizations to evaluate risks with minimal resource investment. It prioritizes critical information assets and includes steps such as defining risk criteria, identifying key assets, identifying threats and vulnerabilities, assessing risks, and creating mitigation plans. Its main advantages are its speed, resource efficiency, and focus on protecting vital business assets.

Recommendation

The recommended risk assessment framework for Fullsoft, Inc. is OCTAVE Allegro because of its streamlined and efficient methodology. This approach prioritizes the protection of critical assets, such as intellectual property, and emphasizes essential business processes while reducing resource expenditures. In comparison, NIST SP 800-30 requires a greater investment of personnel, time, and financial resources, whereas OCTAVE Allegro enables an effective risk assessment with fewer resources.

Justification

Fullsoft, a large software development company that aims to protect its intellectual property. OCTAVE Allegro is recommended as the ideal risk assessment framework due to its ability to safeguard critical assets efficiently and effectively.

Work Cited

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